Hongcheng Jiang

Applied AI Engineer | LLM Agents & Evaluation

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Kansas City, MO | Open to relocation / remote | U.S. Permanent Resident (no sponsorship required)

Summary

Applied AI engineer building tool-using LLM agents with guarded execution, automatic repair, and evaluation. Achieved 91.0% accuracy versus a 65.4% baseline in a text-to-SQL development-set configuration comparison. Ph.D. in Electrical & Computer Engineering; UMKC postdoctoral researcher applying agent patterns to medical measurement.

Technical Skills

Agents: Tool Calling, MCP, Structured Outputs, Guardrails, Bounded Repair, Evaluation Harnesses, Execution Tracing
Engineering: Python, SQL, FastAPI, SQLite, Async Workers, Docker, Linux, Git
ML: PyTorch, Hugging Face, vLLM, Ollama, Computer Vision, Diffusion Models, Vision Transformers

Professional Experience

University of Missouri–Kansas City — Jun 2026 – Present
Research Associate (Postdoc) — LLM Agents & Medical AI — *Kansas City, MO*

- RadMeasure — Medical Measurement Agent: Constrained LLM-proposed actions to registered protocols and tools; implemented geometry verification and a **KEEP/REPAIR/STOP controller** with repair budgets, independent-proposal checks, human review, and execution traces.
- RadMeasure offline study: A learned repair policy selected 106/528 predictions (20.1%) across 176 archived cases; **mean error fell 0.69° among selected records**, with overall MAE decreasing from 2.68° to 2.54°.
- GeoVIRA (submitted, WACV 2027): Conditioned frozen visual features on geometry priors, cutting geometry-only MAE **13.5–25.1%** across three clinical angle tasks (hallux valgus, intermetatarsal, spinal Cobb); isolated the visual contribution with matched-versus-shuffled controls.

NextTier — IT Solutions Consultancy — Jun 2025 – May 2026
Machine Learning Engineer — *Sacramento, CA (Remote)*

- Built SQL-Agent, an **internal prototype** using Qwen3-14B over registered SQLite data, with browser and FastAPI/MCP interfaces for revenue totals, order lists, grouped summaries, and result review.
- Engineered schema-aware **generate–execute–repair** loops with read-only authorization, query/output limits, and up to three attempts on a consistent data snapshot; routed general answers to human review.
- Designed async execution with idempotent jobs, renewable worker leases, retries, and **stale-worker write protection**; verified recovery under worker-kill, restart, duplicate-request, and timeout tests.
- Raised end-to-end accuracy to **142/156 (91.0%) from 102/156 (65.4%)** for a single-pass Coder baseline in a configuration comparison across 26 development questions $\times$ 2 databases $\times$ 3 trials; supported a 74.8 s pipeline p95 with asynchronous delivery.

University of Missouri–Kansas City — Jan 2020 – May 2025
Graduate Research Assistant — *Kansas City, MO*

- Developed transformer- and diffusion-based models for **thermal super-resolution, hyperspectral pansharpening, and NIR-to-RGB translation**; evaluated image quality and spectral fidelity.
- Co-authored a fully connected LSTM-CRF for clinical NER, reaching **84.15% F1 on i2b2/VA**.

Education

University of Missouri–Kansas City
Ph.D., Electrical & Computer Engineering, GPA: 4.0/4.0 — May 2025
M.S., Computer Science, GPA: 3.8/4.0 — Dec 2019

Publications

8 peer-reviewed publications; 6 first-author imaging papers. Full list: Google Scholar.